

Climate and Environmental Intelligence Dashboard

DSA3050A Group 7 Project Report

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1. Project Overview

This project developed an interactive Power BI dashboard to analyze global weather and environmental indicators using the World Weather Repository dataset. The dashboard supports climate monitoring, environmental comparisons, and data-driven decision-making.

2. Business Problem

Global atmospheric volatility and worsening air pollution pose severe risks to municipal public health, city infrastructure, and ecological balance. Environmental protection agencies, public health departments, and urban policy directors often lack a unified, real-time Business Intelligence (BI) platform to continuously monitor spatial-temporal weather trends alongside hazardous air pollutants. Without centralized analytics, identifying vulnerable geographic clusters, tracking dangerous particulate matter accumulation (PM2.5 and PM10), and detecting sudden extreme heat or ultraviolet (UV) radiation spikes remains reactive rather than proactive.

3. Dataset Description

No. of rows:154,556 Rows

No. of columns: 41 columns

No. of tables: 2 tables

This dataset provides daily weather information for capital cities around the world. Unlike forecast data, this dataset offers a comprehensive set of features that reflect the current weather conditions around the world. Starting from August 29, 2023. It provides over 40+ features , including temperature, wind, pressure, precipitation, humidity, visibility, air quality measurements and more. The dataset is valuable for analyzing Global weather patterns, exploring climate trends, and understanding the relationships between different weather parameters.

4. Power Query Transformations

- Renamed unclear columns
- Corrected data types
- Removed duplicates
- Removed blank rows
- Trimmed and cleaned text columns
- Replaced inconsistent values
- Handled missing values
- Removed unnecessary columns
- Split and merged columns
- Created custom columns
- Created conditional columns
- Date_table added
- Summary_Regional_Averages table created

5. Data Model

A star schema was implemented with an Environmental fact table linked to Date, Country, Region, Climate Indicator, Disaster Type (if applicable), and Emission Sector dimensions where appropriate.

6. DAX Measures

Created measures such as Total CO2 Emissions, Average Temperature, Average Rainfall, Average AQI, Average PM2.5, Disaster Count, Water Usage, Forest Loss, Temperature Change, CO2 Growth %, Rank by Emissions, Environmental Risk, YTD Emissions, Countries Monitored, and Dynamic Dashboard Title.

7. Dashboard Pages

Executive Summary, Climate Trends, Geographic & Environmental Analysis, and Detailed Drill-through with KPI cards, maps, line charts, treemaps, scatter plots, matrices, slicers, tooltips, and drill-through.

8. Key Insights

Global temperatures show an upward trend; industrial activity is linked to higher emissions; some regions consistently have poorer air quality; floods and storms are common; deforestation remains significant in selected regions.

9. Recommendations

Reduce emissions in high-impact sectors, strengthen pollution monitoring, expand reforestation, improve disaster preparedness, and promote sustainable water management.

10. Conclusion

The dashboard demonstrates how Power BI, Power Query, and DAX can transform environmental data into actionable intelligence for governments, researchers, and environmental organizations.